

bunch of electrical wires, one small square sheet of transparent acrylic (1mm thick, 8 cm on the side), transparent adhesive tape and heat-sink (optional).

Construct the touchpad grid using bare wires as shown in Fig. 4. Once the touchpad grid wiring is complete, overlay it on the LED matrix (refer Fig. 5). Tape the wires to the acrylic sheet. LEDs are on a separate board underneath this acrylic sheet. So the structure is actually a 'see through' view of the acrylic sheet, showing the LEDs below.

It is important to secure the wire grid firmly in place on a sturdy playing board. Connect toggle switch S1 between digital pin 13 and ground of the Arduino board so that you can

switch between one-player and two-player modes.

The game

Two-player mode. In two-player mode, the tic-tac-toe game proceeds as follows:

Step 1. Player A picks an unfilled cell by touching any of the nine intersection points.

Step 2. The LED for that cell lights up in, say, red colour.

Step 3. Check to see if player A has won.

Step 4. Player B picks an unfilled cell.

Step 5. The LED for that cell lights up in, say, green colour.

Step 6. Check to see if player B has won.

Step 7. Repeat steps 1 through 6

until there is a win condition, or all the cells are filled.

Note that each cell is formed by an LED. A cell is filled when the LED glows. Initially, all cells are unfilled.

One-player mode. If the switch is in the 'on' state (closed), the program goes into one-player mode, with the human player starting first. After the human player's turn, the program itself randomly picks one of the remaining cells.

EFY note. The source code of this article is available on www.efymag.com website.

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TICTACTOE.PDE

```
// tictactoe.pde

#define CountLoop 5 // how many times
// to take a reading
#define Threshold 1010 // sensitivity
// threshold for analog inputs
#define LEDTime 500 // how long the LED
// stays on, in microseconds
#define ModePin 12 // mode selection
// pin
#define ModeLEDPin 13 // which mode: 0
// = 2 player, 1 = 1 player vs uController

int RedLEDPins[] = {9, 10, 11};
int GreenLEDPins[] = {3, 5, 6};
int CathodePins[] = {4, 7, 8};

void setup()
{
  // setting up of mode selection pin
  pinMode(ModePin, INPUT);
  digitalWrite(ModePin, HIGH);
  pinMode(ModeLEDPin, OUTPUT);
  digitalWrite(ModeLEDPin, LOW);
}

// lightLED function lights up the LEDs
// void lightLED(word LEDOnOff, word LED-
// Colour)
{
  // shift the bits to the right, turning
  // on the LEDs whenever
  // there is a 1, turning off whenever
  // there is a 0 in LEDOnOff
  // If the LED is lit, LEDColour deter-
  // mines which LED is lit
  // 1 is red, 0 is green
  for (int j=0; j<3; j++)
  {
    pinMode(RedLEDPins[j], INPUT);
    digitalWrite(RedLEDPins[j], LOW);
    pinMode(GreenLEDPins[j], INPUT);
    digitalWrite(GreenLEDPins[j], LOW);
    pinMode(CathodePins[j], INPUT);
    digitalWrite(CathodePins[j], LOW);
  }

  for (int i=0; i<9; i++)
  {
    if (LEDOnOff & 1)
    {
      if (LEDColour & 1)
      {
        pinMode(RedLEDPins[i/3], OUT-
        PUT);
        pinMode(CathodePins[i%3], OUT-
        PUT);
        analogWrite(RedLEDPins[i/3],
        level);
        digitalWrite(CathodePins[i%3],
        LOW);
        delayMicroseconds(LEDTime);
        digitalWrite(RedLEDPins[i/3],
        LOW);
        digitalWrite(GreenLEDPins[i/3],
        LOW);
        pinMode(GreenLEDPins[i/3],
        INPUT);
        pinMode(CathodePins[i%3], IN-
        PUT);
      }
      else
      {
        pinMode(GreenLEDPins[i/3],
        OUTPUT);
        pinMode(CathodePins[i%3], OUT-
        PUT);
        analogWrite(GreenLEDPins[i/3],
        level);
        digitalWrite(CathodePins[i%3],
        LOW);
      }
    }
    else
    {
      pinMode(RedLEDPins[i/3], OUT-
      PUT);
      pinMode(CathodePins[i%3], OUT-
      PUT);
      digitalWrite(RedLEDPins[i/3],
      LOW);
      digitalWrite(GreenLEDPins[i/3],
      LOW);
      pinMode(GreenLEDPins[i/3],
      INPUT);
      pinMode(CathodePins[i%3], IN-
      PUT);
    }
    LEDOnOff = LEDOnOff >> 1;
    LEDColour = LEDColour >> 1;
  }
}

// lightLED function lights up the LEDs
// void lightLEDPWM(word LEDOnOff, word
// LEDColour, int level)
{
  if (LEDColour & 1)
  {
    pinMode(RedLEDPins[i/3], OUT-
    PUT);
    pinMode(CathodePins[i%3], OUT-
    PUT);
    analogWrite(RedLEDPins[i/3],
    level);
    digitalWrite(CathodePins[i%3],
    LOW);
    delayMicroseconds(LEDTime);
    digitalWrite(RedLEDPins[i/3],
    LOW);
    digitalWrite(GreenLEDPins[i/3],
    LOW);
    pinMode(GreenLEDPins[i/3],
    INPUT);
    pinMode(CathodePins[i%3], IN-
    PUT);
  }
  else
  {
    pinMode(GreenLEDPins[i/3],
    OUTPUT);
    pinMode(CathodePins[i%3], OUT-
    PUT);
    analogWrite(GreenLEDPins[i/3],
    level);
    digitalWrite(CathodePins[i%3],
    LOW);
  }
}
```